

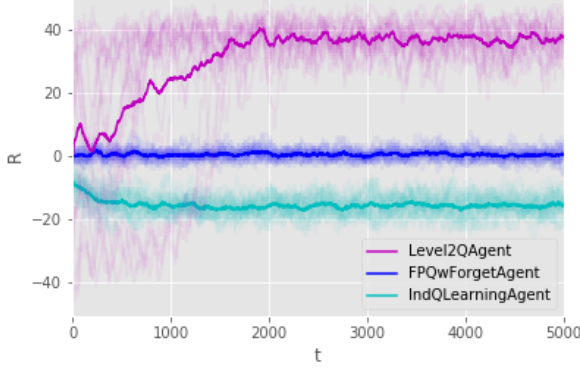


Figure 8: Rewards against the adversarial opponent

DM with forget and a level-2 agent. The first one is exploited by the adversary and, therefore, achieves suboptimal results. In contrast, the level-1 DM with forget effectively learns an stationary optimal policy (reward 0). Finally, the level-2 agent learns to exploit the adaptive agent achieving positive reward.

Note that the actual adversary behaves differently from how the DM models him, i.e. he is not exactly a level-1 Q-learner. Even so, modeling him as a level-1 agent gives the DM sufficient advantage.

Spatial Variant We now compare the independent Q-learner and a level-2 Q-learner against the same adaptive opponent in a spatial gridworld domain, Figure ?? . Targets’ rewards are delayed until the DM arrives at one of the respective locations, obtaining ± 50 depending on the target chosen by the adversary. Each step is penalized with a reward of -1 for the DM. Results are displayed in Figure 9. Once again, the independent Q-learner is exploited by the adversary, getting even more negative rewards than in Figure 8 due to the penalty taken at each step. In contrast, the level-2 agent is able to approximately estimate the adversarial behavior, modeling him as a level-1 agent, thus being able to obtain positive rewards.

4 Conclusions and further work

We have introduced TMDPs, a novel variant of MDPs. This is an original framework to support decision makers who confront adversaries that interfere with the reward generating process in reinforcement learning settings. TMDP’s aim to provide one-sided prescriptive support to a DM, maximizing her subjective expected utility, taking into account potential negative actions taken by an adversary. Some theoretical results are provided, in particular, we proved that our proposed learning rule is a contraction mapping so that we may use standard RL results of convergence. In addition, we propose a scheme to model adversarial behavior based on level- k reasoning about opponents. Further empirical evidence is provided via extensive experiments, with encouraging results.

Several lines of work are possible for further research.

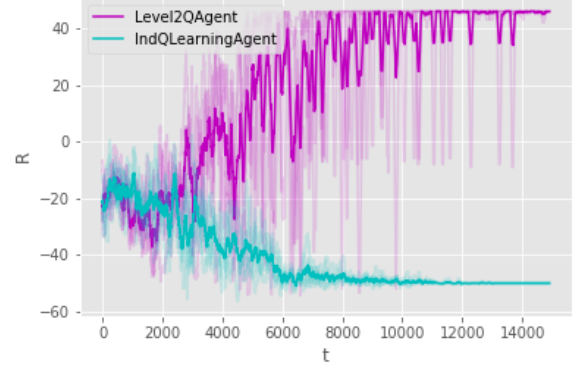


Figure 9: Rewards against the adversarial opponent in the spatial environment

First of all, we have limited to the case of facing just one adversary. The framework could be extended to the case of having multiple adversaries. In the experiments, we have just considered up to level-2 DM’s, though the extension to higher order adversaries seems straightforward.

In addition, in recent years Q-learning has benefited from advances from the deep learning community, with breakthroughs such as the *deep Q-network* (DQN) which achieved super-human performance in control tasks such as Atari games (?), or as inner blocks inside systems that play Go (?). Integrating these advances into the TMDP setting is another possible research path. In particular, the proposed Algorithm 1 can be generalized to account for the use of deep Q-networks instead of tabular Q-learning.

Finally, it might be interesting to explore similar expansions of semi-MDPs, in order to perform Hierarchical RL or allow for time-dependent rewards and transitions between states.

5 Acknowledgments

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